

1. We may formulate the force from a constant magnetic field on a current carrying loop with constant current as

$$\vec{F}_m = I \oint_C d\vec{l} \times \vec{B} = Il \times B.$$

Finding the force on each side of the triangle formed by the current carrying loop, beginning with the left side and proceeding clockwise,

$$\begin{aligned}\vec{F}_1 &= 10^A(\hat{x}(0.1) + \hat{y}(0.2))^m \sin(-26.56^\circ) = \hat{z}(6)^{mN} \\ \vec{F}_2 &= 10^A(-\hat{x}(0.1) + \hat{y}(0.2))^m \sin(26.56^\circ) = \hat{z}(6)^{mN} \\ \vec{F}_3 &= 10^A(\hat{x}(0.2))^m \sin(90^\circ) = -\hat{z}(12)^{mN}.\end{aligned}$$

Deriving the magnetic moment of the single loop of wire using the current and area of the loop,

$$\vec{m} = \hat{n}IA = -\hat{z}[10^A(0.2^m)]^{A \cdot m^2}$$

and the corresponding torque,

$$\vec{T} = \vec{m} \times \vec{B} = 2(0.006) \sin(90^\circ) = -\hat{x}(12)^{mN \cdot m}.$$

2. We may formulate the magnetic flux density $\vec{B}$ from an infinitely long wire as

$$\vec{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r},$$

with the direction of $\hat{\phi}$ determined by the right-hand rule.

In this case, the magnetic flux density of the two current carrying wires being considered is

$$\begin{aligned}\vec{B}_1 &= \hat{\phi} \frac{\mu_0(10^A)}{2\pi(\sqrt{8})}^T \\ \vec{B}_2 &= -\hat{\phi} \frac{\mu_0(10^A)}{2\pi(\sqrt{8})}^T.\end{aligned}$$

Deriving force on the third wire as in the previous example,

$$\vec{F}_3 = I \int_L d\vec{l} \times \vec{B} = 10^A \int_L d\hat{z} \times (\vec{B}_1 + \vec{B}_2).$$

Converting the magnetic flux densities to Cartesian coordinates,

$$\begin{aligned}\vec{B}_1 &= [-\hat{x}(5 \times 10^{-7}) + \hat{y}(5 \times 10^{-7})] \\ \vec{B}_2 &= [\hat{x}(-5 \times 10^{-7}) - \hat{y}(5 \times 10^{-7})]\end{aligned}$$

so that

$$\vec{F}_3 = 10^A \int_L d\vec{z} \times \hat{x}(10^{-6}) = \hat{y}(10)^{mN}$$

3. Using Ampere's Law,

$$\oint_C \vec{H} \cdot d\vec{l} = I,$$

which states that the line integral of a magnetic field $\vec{H}$ around a closed path is proportional to the current flowing through the surface area enclosed, we may express the relationship in this case, for $0 < r < a$, (using the right-hand rule to find the direction of $\vec{H}$), as

$$2\pi rH = \frac{J_0}{r}(\pi r^2) \Rightarrow \vec{H} = \hat{\phi} \left(\frac{J_0 r}{2} \right) \frac{A}{m}.$$

For $r > a$,

$$\vec{H} = \hat{\phi} \left(\frac{J_0 a^2}{2r^2} \right) \frac{A}{m}.$$